

MEAN FROM FREQUENCY TABLES

Foundation Tier

Mean and estimated mean

Name

Date

8 questions

26 marks

2017-2025

Instructions

- Answer each question in the space provided.
- Show all your working and follow the rounding instructions.
- Diagrams are not accurately drawn unless stated otherwise.
- Use a calculator except for pages marked NON-CALCULATOR.

Using this booklet

Table questions are arranged by exam series, followed by any chart or histogram extensions. Each page gives its original paper reference and question number. The question numbers within the extracted artwork are the original exam numbers.

Only the specified question parts are included. Shared questions appear in both tier booklets so that each can be used independently. Original question numbers and part labels are retained.

This is an independently compiled practice resource, not an official Pearson examination paper. The source collection covers standard UK June and November papers from 2017-2025. No solutions are included.

Question index

Foundation Tier | 8 questions | 26 marks

Practice	Exam series	Paper	Original question	Marks	Page
01	November 2017	1MA1/1F	Q27	4	3
02	November 2019	1MA1/3F	Q25	3	4
03	November 2022	1MA1/2F	Q17	3	5
04	November 2023	1MA1/3F	Q22(a)	3	6
05	June 2024	1MA1/2F	Q28(b)	3	7
06	November 2024	1MA1/3F	Q15(a)	3	8
07	June 2025	1MA1/3F	Q19(b)	3	9
08	November 2018	1MA1/3F	Q23(b)	4	10

The original exam numbering is retained in each question.

Paper 1 questions are marked NON-CALCULATOR on their pages.

Chart / histogram extensions: practice 8.

27 The table shows information about the weekly earnings of 20 people who work in a shop.

Weekly earnings (£ x)	Frequency
$150 < x \leq 250$	1
$250 < x \leq 350$	11
$350 < x \leq 450$	5
$450 < x \leq 550$	0
$550 < x \leq 650$	3

(a) Work out an estimate for the mean of the weekly earnings.

£.....
(3)

Nadiya says,
“The mean may **not** be the best average to use to represent this information.”

(b) Do you agree with Nadiya?
You must justify your answer.

.....
.....
(1)

(Total for Question 27 is 4 marks)

25 The table gives information about the times taken, in seconds, by 18 students to run a race.

Time (t seconds)	Frequency
$5 < t \leq 10$	1
$10 < t \leq 15$	2
$15 < t \leq 20$	7
$20 < t \leq 25$	8

Work out an estimate for the mean time.
Give your answer correct to 3 significant figures.

..... seconds

(Total for Question 25 is 3 marks)

Additional working space

Height (h cm)	Frequency
$150 < h \leq 160$	8
$160 < h \leq 170$	14
$170 < h \leq 180$	24
$180 < h \leq 190$	30
$190 < h \leq 200$	4

Work out an estimate for the mean height of the teenagers.

..... cm

(Total for Question 17 is 3 marks)

Additional working space

22 Seija works at a weather station.
The table gives information about the temperature, T °C, at midday for each of 50 cities in the UK on Tuesday.

Temperature (T °C)	Frequency
$10 < T \leq 15$	2
$15 < T \leq 20$	8
$20 < T \leq 25$	13
$25 < T \leq 30$	21
$30 < T \leq 35$	6

(a) Calculate an estimate for the mean temperature.

..... °C

(3)

Additional working space

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.....

28 The table shows information about the weights of 120 oranges.

Weight (w grams)	Frequency
$50 < w \leq 100$	34
$100 < w \leq 150$	29
$150 < w \leq 200$	27
$200 < w \leq 250$	19
$250 < w \leq 300$	11

- (b) Calculate an estimate for the mean weight of the 120 oranges.
Give your answer correct to 3 significant figures.

..... grams

(3)

Additional working space

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.....

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Age (years)	Frequency
8	6
9	7
10	15
11	11
12	2

- (a) Work out the mean age.
Give your answer correct to 1 decimal place.

..... years
(3)

19 Katya counted the number of people in each of 50 cars.

The table gives information about her results.

Number of people	Frequency
1	16
2	15
3	7
4	9
5	3

(b) Work out the mean number of people.

.....
(3)

Additional working space

.....

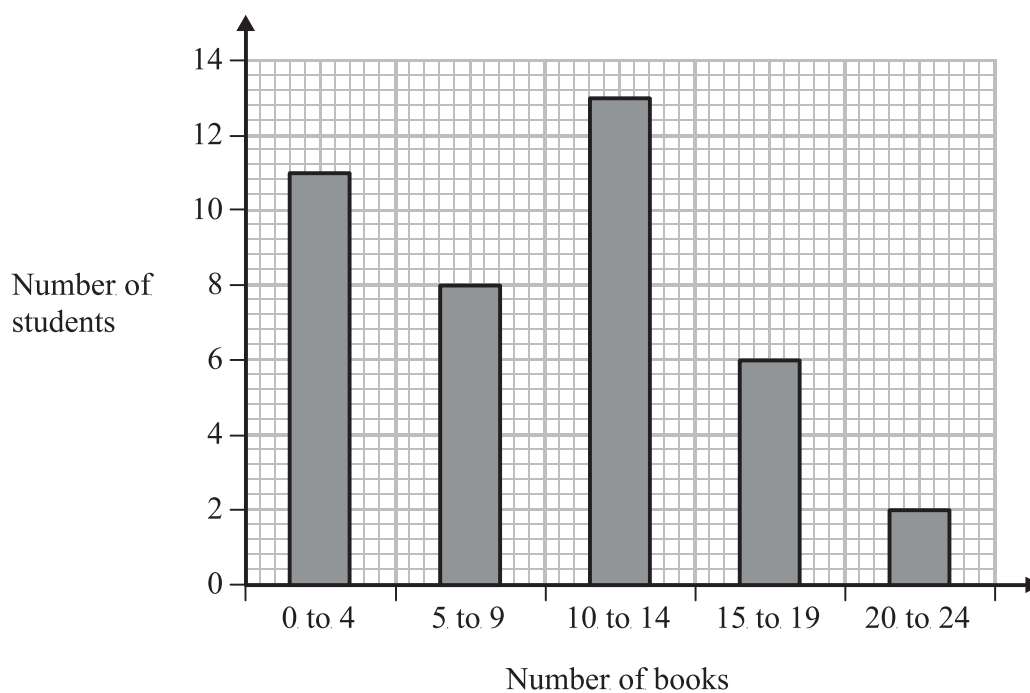
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23 Fran asks each of 40 students how many books they bought last year.

The chart below shows information about the number of books bought by each of the 40 students.



- (b) Show that an estimate for the mean number of books bought is 9.5
You must show all your working.

(4)